

Koa Shen

Mechanical Engineering Intern Candidate · Equipment Design · Analysis · Manufacturing

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EDUCATION

University of California, Santa Barbara

Expected June 2028

B.S. Mechanical Engineering · 3.97 major / 3.94 overall GPA · Dean's Honors every quarter

Santa Barbara, CA

- Coursework: Dynamics, Mechanics of Materials, Circuits. Current: Dynamical Systems and Machine Learning.

EXPERIENCE

Mechanical Engineering Intern

Summer 2026 – Present (extended to part-time remote)

SLAC National Accelerator Laboratory — LCLS

Menlo Park, CA

- Designed and analyzed high-DOF X-ray beamline assemblies using CAD, Python/Drake kinematics, collision checks, and load cases.
- Produced Solid Edge designs with GD&T and Teamcenter documentation; used analysis findings to guide optics assembly redesigns.

Fabrication Team Lead

Fall 2024 – Present

Gaucha Racing (FSAE EV) — UC Santa Barbara

Santa Barbara, CA

- CNC-machined 33 components from 4130 steel and 7075-T6 aluminum; held critical bearing interfaces within 0.0005 in.
- Led 15 teammates through safety-system design, manufacturing, and installation; passed FSAE technical inspection.

Maintenance Technician

Summer 2025

Exploratorium

San Francisco, CA

- Maintained 100+ exhibit mechanisms, reverse-engineered and fabricated replacement components, and led three major overhauls.
- Redesigned a cracking acrylic enclosure interface with a load-spreading bezel, avoiding \$3,000+ in replacement costs.

SELECTED PROJECTS

Digital Twin Simulation Framework for LCLS Motion Stages

2026 – Present

- Built a Python/Drake digital twin for a 32-DOF assembly, integrating CAD import, kinematics, collision checking, and visualization.
- Applied the framework to three LCLS assemblies; collision checks run in milliseconds across 305 bodies.

Polycapillary Optics Assembly Redesign

2026 – Present

- Redesigned a high-DOF optics assembly with three kinematic bases, six hot-swappable optics, and added motion stages.
- Used simulation findings, instrument-scientist feedback, and statics/dynamics checks to guide packaging and alignment changes.

GR26 CNC Wheel Assemblies

Winter 2025 – Spring 2026

- CNC-machined 33 precision suspension parts on Haas mills and lathes; held critical bearing fits within 0.0005 in.

TECHNICAL SKILLS

CAD & Engineering Documentation: Solid Edge, SolidWorks, Teamcenter PDM, GD&T, Engineering drawings

Analysis & Simulation: Statics and dynamics, FEA, Python, MATLAB, Drake kinematics, Collision detection

Manufacturing & Hardware: CNC/manual machining, Mastercam, 4130 steel, 7075-T6 aluminum, Welding, Design for manufacturing